

Professional Experience Summary

- Having 20+ years of experience in the field of software development, maintenance, support and Testing.
- Around 10 years of experience in Project management and Program management
- Main area of experience in Requirement gathering, GUI development and design of windows desktop applications
- Experience in programming skills – VC++, MFC, InstallShield and Advanced Installer
- Good Knowledge of Windows OS, Windows SDK
- Good experience in InstallShield and Advanced Installer packaging, VB Scripting and Service-pack installations.
- Experience in designing, developing and using DLLs and Libraries.
- Working experience with Windows services, Windows Registries and Windows Group policy
- Knowledge of SQL Server Database design, Database maintenance, developing SQL queries
- Effective communication with excellent negotiation skills, leadership qualities and analytical power.
- Involved in High Level design such as Approach Notes, Feasibility study, POC
- Thorough understanding of the SDLC with proficiency in mapping client/business requirements, documentation, application design, development, integration & testing and troubleshooting for complex information systems management.
- Proven ability to learn/adapt emerging technologies into work area with given rigid schedules.
- Knowledge about ISO procedures and TQM
- My current role is Technical Manager for application product development for thin client windows environment. Study of competitor product and inline development of our product to industry requirement level.
- Lead a team of 10 developers
- Responsible for complete end to end development along with support, including adhoc requests.
- Full responsibility of technical reviews, documentation and client communication.
- Coach/mentor colleagues in design and solve day to day technical issues.
- Agile methodologies using Scrum/Kanban
- Working experience and knowledge of Software Design Methodologies, Object Oriented Design, Software Design and Architectural Patterns.
- Working knowledge of SOLID Design Principles, Design Pattern, STL, Data structures, Algorithms, Socket programming, COM and Multithreading.
- Implemented Unit Test cases
- Fix the Security flaw issues identified by Coverity / Veracode Static Analyzer tool
- Responsible for fixing issues from Penetration testing (Vulnerabilities) and take them to the closure

Work Profile:

- Currently working with **VXL Instruments Ltd. Bangalore** as a **Technical Manager** from July 2007 – June 2024
- Worked with Indo-Fuji Information Technologies Private Ltd. Bangalore as a Project Leader from Aug 2003 – July 2007
- Worked with Infotech-PRDC Private Ltd. Bangalore as a Software Engineer from Jan 1999 – Aug 2003.

Qualification:

B.E. Computer Science & Engg. B.I.T., Bangalore.

Technical Skills:

Langages

C, C++, SQL

Micorsoft Technologies

VC++, MFC, Win32 API, COM, STL, C#, WPF

Databases

Microsoft SQL Server 2012

Operating Systems

Unix, Linux, Windows 10

IDE
Tools

Microsoft Visual Studio 2012

Visual Source Safe, Tortoise SVN Client, Bugzilla, Install shield, Advanced Installer

Onsite-Experience:

Onsite : Seoul – South Korea

Company : Nitgen Co., Ltd.

Project : Fingerprint Authentication Subsystem of Server Applications

Purpose : Onsite co-ordination for complete requirement gathering, making design document and review all phases of SDLC with customer till the Testing and Delivery of the product.

Onsite : Kyoto – Japan

Company : ARTICK

Project : VT-RNS

Purpose : Development of the assigned modules using VC++ and MFC

Domain Knowledge:

- Thin client – Remote Management, Windows Embedded, Windows Custom Applications
- Industrial Automation

Projects Profile:

1. Title: Thin-client Applications

Period: From Sept 2015 to till date

Tech Env: C#, WPF

Description: Involved in the Thin client application development / modification explained each here below.

Role:

- Involved in Analysis, Requirements gathering, Development co-ordination, Testing.
- Review of designed GUI using C#, Win-Form, WPF.
- Review of business logic in C#.
- Performed unit testing and co-ordination.

Module title: Auto-logon Manager

Description: By default, a thin client is automatically logged on to the locked-down User account. The administrator can use the Auto-Logon Manager to enable/disable auto logon and change the auto logon user name, password, and domain. Only the Administrator has the privileges to change auto logon properties. Automatic Log On allows a user to log on without entering the user name and password. This type of log on is useful when using the thin client as a public access computer or any computer that does not require user access rules.

Module title: Performance Controller

Description: Performance Monitor and Warning Settings provides a visual display of real-time information about hardware resources (CPU, Network, Memory and Disk). In addition the user can set Warning levels. Utilizations are updated in regular interval of time. Notifications will popup if any of the monitor utilization exceeds warning level once warning setting is enabled.

Module title: USB Device Controller

Description: USB Device Controller helps the user to enable as well as disable peripherals which uses interface as USB. After reboot the peripherals which are disabled will not get detected even if it is connected.

Module title: UWF Manager

Description: Unified Write Filter (UWF) manager is a client application to manage the filter for Windows Embedded 8 /10. It is used to protect storage media. UWF intercepts all write attempts to a protected volume and redirects those write attempts to a virtual overlay.

Module title: Configuration Manager

Description: It's an application, generates configuration file and configures the thin clients based on generated configuration file. It consists of UI application and windows service. It supports ftp, http and https protocols.

Module title: Win 10 IoT MUI Installer

Description: It's an Install-shield package application package, which will be installed on Win10 IoT English image and apply all the required changes to take effect for a particular MUI language.

Module title: Win 10 IoT Security Patch Installer

Description: It's a Generic Install-shield application package, validates and Installs the Microsoft Security updates/QFEs on Win 10 IoT.

2. Title: VXL Connection Launcher (VCL)**Role:**

- Involved in Analysis, Requirements gathering, Development co-ordination, Testing.
- Review of designed GUI using C#, WPF.
- Experience in development connecting from Windows Shell mode to VDI apps like Citrix, VMware, MSRDC, AWS
- Review of business logic in C#.
- Performed unit testing and System testing co-ordination.

Description: VXL Connection Launcher (VCL) is an application that is used to configure Windows based terminals and other supported devices with a kiosk type environment that allows easier access to virtual desktop connections. VCL enables administrative users to configure and control connections such as Citrix, VMware, RDP, WVD, AWS, Browsers Chrome and Edge while also linking them to user profiles.

The VCL enables the administrators to create a self-extracting executable file which can be deployed to Win10 IoT thin clients, as well as other supported Windows devices.

The administrator, using VCL, can automate the log on experience for users so that end-user interaction is minimized. VCL configures connections within the thin client based on the user logon information detailed within the .INI file.

3. Title: Imaging Utility**Role:**

- Involved in Analysis, Requirements gathering, Development co-ordination, Testing.
- Involved in customizing Windows Pre Environment OS and creating/modifying boot order using Microsoft commands.
- Review of designed GUI using C#, WPF.
- Review of business logic in C#.
- Involved in creating Imaging Utility Package using Advanced Installer
- Performed unit testing and co-ordination.

Module Title: Cedar

Description: Cedar is a utility developed for disk cloning or disk imaging locally and as well from remote over CIFS Share and ftp platform. Cedar create a new boot entry for the customized Windows Pre Environment along with the existing windows OS and helps to you to do system deployment, backup and recovery operations.

Module Title: USB Imaging Utility

Description: USB Imaging Utility is a utility developed for disk cloning or disk imaging locally and as well from remotely over the network through USB disk. It prepares the bootable Widows Pre environmental platform on the USB. Windows OS boots from the USB and helps you to do system deployment, backup and recovery operations.

4. Title: Image Locker

Involved in the Image Locker requirement gathering, development/ modification and enhancement coordination.

Role:

- Involved in Analysis, Requirements gathering, Development co-ordination, Testing.
- Review of designed GUI using C#, WPF and windows service.
- Review of business logic in C#.
- Involved in creating Image Locker Package using Advanced Installer
- Performed unit testing and system testing co-ordination.

Description: Image Locker is a utility developed for restricting Windows OS usage from unauthorized access by providing evaluation period for the users. After the evaluation period the Windows Operating Systems becomes unusable state. Customer need to buy an activation key to get uninterrupted usage of Windows Operating System.

5. Title: PC to TC Converter

Involved in the Personal Computer to Thin Client Converter for Windows 10 Enterprise Operating System requirement gathering, development/ modification and enhancement coordination.

Role:

- Involved in Analysis, Requirements gathering, Development co-ordination, Testing.
- Review of designed GUI using C#, WPF and windows service.
- Review of business logic in C#.
- Involved in creating PC to TC Package using Advanced Installer
- Performed unit testing and system testing co-ordination.

Description: PC to TC Converter is a software utility provides a thin client solution with existing OS, reducing risk and lowering the cost of hardware. It provides optimal thin client user experience with simple software bundle for Windows 10 Enterprise Operating System. It enables the UWF features, USB Device Controller and adds Windows Connection Launcher experience and updating VDI features.

6. Title: XLmanage–A thin-client remote management Application

Period: From July 2007 to Aug 2015 date

Tech Env: MFC, VC++, SQL Server 2005 / 2008

Description:

XLmanage is a remote management suite that allows the System Administrator to efficiently control and configure VXL Thin Clients. This application is a standard feature of VXL Thin Client solution. The utility is especially valuable in managing multiple clients in large enterprises with terminal services like Citrix ICA, Microsoft RDP services.

Role: As a member of core development team, I was involved in design and development of GUI, Database and unit testing of new features added to existing application.

Module title: XLmanage MUI – Chinese

As a member of core development team, I was involved in design and development of concept implementation for complete product GUI into Chinese. Product was redesigned from Multi-Byte character set to Unicode character set for MUI. Design implemented in such way that a new DLL for each MUI language to change the English strings into Chinese using string table concept. Depending on the Language selection corresponding Unicode DLL was loaded to achieve the MUI support.

Module title: User management Integration with Active Directory

Importing of selected AD users / Groups / OUs into XLmanage User management with default or configured privileges. These AD users are allowed to login-to XLmanage application with AD authentication at login-time. For each user, a group can be assigned with privileges which are configured by administrator.

Module title: USB Access control in Thin client

Implemented a Remote USB access level control module to restrict the usage of a USB device in thin client. A schedule is required to enable the access control feature at client side. When a USB device is plugged-in at thin client automatic request will be sent to server application to give the access rights. Access rights can be USB Read-only, Read-write and Deny which will be decided by administrator for each/multiple thin-clients request at any point of time.

Module title: Sub-grouping

Sub-grouping was implemented based on the Grouping criteria IP Range, Subnet and Custom. Each group/Sub-group can be attached with pre-configured profile, which can be modified depending on the usage. Automatically a profile will be scheduled to the client when it falls into any configured grouping criteria.

Module title: Join/Unjoin to Domain (WES7 Clients)

This is type of a schedule sent from XLmanage server to thin client for join/un-join Active directory for Windows 7 thin clients.

Module title: Reporting module

Implemented reporting module to export the detailed Client information and assert information to .csv file, based on the required information fields selected by administrator.

Module title: Configuration Tool

Implemented Configuration Tool which is responsible to configure (Get and Set) the client completely.

Module title: Product installer using Install shield

Worked in Product installer preparation using Install shield 7.0 and automated settings of Database, IIS, COM Components and Firewall.

Database – Worked on attaching and detaching database using scripts.

IIS – Creating custom web site along with application pool using scripts

COM components – Registering com components and setting the Identity using scripts

Firewall – Adding required application files to firewall exception list using scripts.

Module title: Product Service-pack installer

Worked in Product Service-pack installer for upgrading existing version of XLmanage into latest version which is developed in VC++ / MFC.

Process Documentation:

Responsible in ISO related project complete life cycle process documentation.

7. Title: VT-RNS

Period: From April 2006 to Aug 2006

Tech Env: MFC, VC++

Client: OMRON

Description:

The VT-RNS is a system designed to inspect the printed conditions of cream solder and the mounted conditions of parts automatically. The inspection process employs unique image processing techniques specially designed for the color highlight system. This uses a three colours, LED configuration and full colour camera for detailed and clear imaging. Its robust design and multi-language graphical user interface minimize training time and make it easy for operators to control. It is possible to output the inspection results to either the monitor or printer.

Role: As a member of development team, I was involved in coding and testing of new features added to existing application. I was involved in DLL development of Image capturing from USB camera. I have developed Image tool application with GUI interface for testing of image capturing, setting various camera parameters.

8. Title: Fingerprint Authentication Subsystem of Server Applications

Period: From Feb 2005 to Feb 2006

Tech Env: MFC, VC++, C++, CORBA, Web Services, SQL Server

Client: NITGEN Co., Ltd, Korea

Description:

The objective of project is to develop a Finger Print Authentication system that can be customized easily for 1:N and 1:1 systems and authenticating ways such as 1:N and 1:1 combined and developed using eNBS SDK APIs. The system aim at providing improved performance, network security, capability and interoperability with ease. The technologies used are: Java, JSP, Servlet, EJB, MFC, C++, Corba and Web Services. The client application can be MFC or JSP, which interacts with

the EJB using Web services or http. The EJB in turns communicates with the C++ Search Servers through Corba and returns the relevant data.

Responsibility: I was **onsite coordinator** for this Project. My roles involve understanding client requirements preparation of detailed requirement specification document. If any queries from offshore team, getting clarification for the same. Involved in review with the client for Functional design, Structural design, Coding and testing reviews. After execution of the project testing is done at client place and coordinated for acceptance testing.

9. Title: PC Cluster Monitoring Tool

Period: From May 2004 to Nov 2004

Tech Env: Linux 9.0, PERL, CGI, C

Client: Fujitsu, Japan

Description:

The objective of project is to develop a monitoring tool for cluster of Linux servers. Cluster of Linux servers consists of multiple IA servers and a Management server. The tool will send the request from Management server node to other IA Servers in the cluster to get the information about IA servers and itself like generic information, hardware information, software information and history information and this information is provided in web interface also.

As a team lead in the project, involved in functional design, structural design and coding of Client/Server programs for communication between Management server and IA servers using sockets. Multithreading is used to send request simultaneously to all IA servers and getting back the information. Also responsible in ISO related project process documentation.

10. Title: WINCAT 2.1

Period: From Aug 2003 to Apr 2004

Tech Env: Windows XP, MFC/VC++, MS-Access

Client: System Japan, Japan

Description:

WINCAT is the software for Automobile industries to manage all the aspects of a company. WINCAT 1.0 was developed for old version of windows using VC++4.0. Now, We are upgrading this software to Windows XP. Some additional features are incorporated with old version with modification of Functional Design. The critical part of this project is compromising with their program structure, design and coding convention.

As a core member of development team, I was involved in design and development of enhancements of the product and White box testing of the enhanced modules.

11. Title: Automatic Chassis Marking System.

Period: From Dec 2002 to Aug 2003

Tech Env: Windows 98, MFC/VC++ 6.0, MS-Access

Description:

A vehicle that is manufactured by Ashok Leyland has Engine number and Chassis number. Ashok Leyland wants to print Chassis No and the Engine number on the chassis of the vehicle. The number to be punched is pre-scheduled by the supervisor. Operator has to select Chassis type and Chassis number in both modes, namely Chassis Mode and Engine & Chassis Mode. In Engine & Chassis mode, the operator will enter corresponding Engine number. Start print command will be given by the push-button.

The system consists of the TMC 400 Controller. The marking system software has to communicate to TMC 400 controller about the marking functionality. Control various elements in real time through Optoisolated digital IO card to PLC, and local book keeping for report generation and accountability. MS Access used as back end storage.

As a core member of development team, I was involved in design and development and testing of this module. **I was able to develop good GUI interface** for Operator and Supervisor mode. MSComm control is used for communication with TMC 400 Controller. Multithreading concept is used to know the status of the printing or aborted signal. ODBC interface is used for accessing backend database.

12. Title: Custom Report Writer.

Period: From May 2002 to Oct 2002

Tech Env: Windows 98, MFC/VC++ 6.0, MS-Access

Description:

MiPower is a user-friendly windows based power system analysis package, which is a product developed at PRDC. It includes a set of programs for performing power system analysis and design. The customized

report writer is an add-on module for creating customized reports from the data available in MiPower centralized database.

I was involved in design and development and testing of this module. The customized report writer software reads from a database file (mdb file) or from ODBC data source and selects all the tables in that database or data source and all the fields in each table. User can select any required fields for the report from different tables. Report can be sorted based on any report field in any order like ascending, descending or original order. Any field can be selected for finding Sum / Avg. / Max / Min. Reports can be displayed based on the query which is set by the user. I have used serialization technique to store the report templates. I have designed this module to work for any database.

13. Title: Cable Ampacity

Period: From Nov 2001 to Apr 2002

Tech Env: Windows 95, MFC/VC++ 5.0, MS-Access

Description:

Cable Ampacity is designed to simulate the thermal behavior of power cable installations. This program is capable of performing steady state analysis for a wide variety of power cable installations. In order to achieve that with the maximum possible efficiency & flexibility a cable library manager to accommodate the various cable type designs are provided. This program has done in confirmation with Neher-McGrath Method.

14. Title: Computer Aided Design Of Cranes (CraneD).

Period: From May 2001 to Oct 2001

Tech Env: Windows 95, MFC/VC++ 5.0, MS-Access

Description:

CraneD is a design automation software developed to cater to the needs of crane manufacturers. The software is in conformation with IS 3177:1999 and IS 3938:1983. The different modules supported are Hoist, Cross Travel, Long Travel, Slew and Structural. The specification covers Mechanical, Electrical and Structural designs. The different reports generated are approval, design, purchase, technical and clearance diagram documents. It has the following facilities

- ☐ Interactive input selection
- ☐ Database management and updation & Administrative security
- ☐ Output presentation and organization in RTF format
- ☐ Context sensitive help
- ☐ Optional module selection

15. Title: Automation of ID Plate Marking System

Period: From May 2000 to Feb 2001

Tech Env: Windows 95, MFC/VC++ 5.0, MS-Access

Description:

The ID Plates for the Cars manufactured by Maruti Udyog Ltd. are to be marked, which are fixed to the body of the car. The ID plate contains the Engine Number, Chassis Number, Model Code and Date. During marking only the chassis of the vehicle is pre assembled but not the engine. Hence engine details are not available as it is going to be fixed at later stage of assembly.

The marking system must obtain the matched engine number from the central PPC host and then mark it on the ID plate. The system consists of the TMC 400 Controller, Barcode scanner, IO interface unit and a central Host. The marking system software has to communicate to TMC 400 controller about the marking functionality, communicate to host to obtain the engine number and other functionality, control various elements in real time through IO interface unit and local book keeping for report generation and accountability.

16. Title: Relay Characteristics Plotting Software

Period: From Jan 2000 to April 2000

Tech Env: Windows 95, MFC/VC++ 5.0, AutoCAD R13

Description:

The plotting software reads the time stamped impedance file created by SIMPO package and plots the values on the impedance plane. The selected relay characteristics are super imposed on the plotted relay impedance loci.

The input file is in ASCII format and output file is generated in DXF format compatible to Autocad Release-13. Plot contains title of the plot, relay characteristics, impedance plot and time in each zone. Context sensitive help is provided to use the package.